

SECTION 07 95 00

EXPANSION JOINTS

02/09

AMENDMENT 009, 03-16-2026

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1399/E1399M	(1997; R 2022) Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems
ASTM E 84	(2018) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E1612	Standard Specification for Preformed Architectural Compression Seals for Buildings and Parking Structures

UNDERWRITERS LABORATORIES (UL)

UL 2079	(2020) Tests for Fire Resistance of Building Joint Systems
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1.2 SYSTEM DESCRIPTION

1.2.1 General Requirements

Provide expansion control systems of design, basic profile, materials, and operation indicated. Provide units with capability to accommodate variations in adjacent surfaces.

- a. Furnish units in longest practicable lengths to minimize field splicing. Install with hairline mitered corners where expansion control systems change direction or abut other materials.
- b. Include factory-fabricated closure materials and transition pieces, T-joints, corners, curbs, cross-connections, and other accessories as required to provide continuous expansion control systems.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00.00 22 SUBMITTAL PROCEDURES (PWD ME):

SD-01 Preconstruction Submittals

Product Schedule; G

SD-02 Shop Drawings

Expansion Joint Shop Drawings; G

SD-03 Product Data

Interior Expansion Control Systems; G

Floor-to-Wall; G

Wall-to-Wall; G

Aluminum Joint Covers; G

Fire-Resistance Rating; G

SD-06 Test Reports

Product Test Reports; G

SD-07 Certificates

Installer Qualifications; G

1.4 QUALITY ASSURANCE

1.4.1 Installer Qualifications

Engage an experienced installer that is certified, licensed, or otherwise qualified by the expansion joint manufacturer as having the necessary experience, staff, and training to install the manufacturer's products in accordance with specified requirements. Each installer of expansion joint material must be trained, have a minimum of 3 years experience and a minimum of three installations using expansion joint of the type specified. A manufacturer's willingness to sell its products to the Contractor or installer does not infer qualification of the buyer.

1.4.2 Pre-Installation Meeting

Hold a meeting with the installer, field testing agency, the manufacturer, subcontractors (whose employees come into contact with the expansion joint), and the Contracting Officer prior to the installation of any expansion joint material to review the substrates for acceptability, method of application, applied thickness, patching, repair, inspection and testing procedures.

1.5 EXPANSION JOINT SHOP DRAWINGS

For each expansion control system specified provide plans, sections, details, splices, blockout requirements, and attachments to other work.

1.6 PRODUCT SCHEDULE

Prepared by or under the supervision of the supplier. Include the following information in tabular form.

1. Manufacturer and model number for each expansion control system.
2. Expansion control system location cross-referenced to

- Drawings.
- 3. Nominal joint width.
- 4. Movement capability.
- 5. Classification as thermal.
- 6. Materials, colors, and finishes.
- 7. Product options.
- 8. Fire-resistance ratings.

1.7 PRODUCT TEST REPORTS

For each fire/smoke barrier provided as part of an expansion control system, for tests performed by a qualified testing agency.

1.8 DELIVERY, STORAGE, AND HANDLING

Packaged material must be delivered in the original unopened containers, marked to show the brand name, the manufacturer, and the UL markings. Fireproofing material must be kept dry until ready to be used, and must be stored off the ground, under cover and away from damp surfaces. Damaged or opened containers will be rejected. Material with shelf-life must be applied prior to expiration of the shelf-life.

1.9 PROJECT/SITE CONDITIONS

1.9.1 Temperature

Maintain substrate and ambient air temperatures must be maintained within the limits recommended by the expansion joint manufacturer.

PART 2 PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

Sealer must be the type approved by the manufacturer of the expansion joint material, must be fungus resistant, must have a flame spread of 25 or less and a smoke developed rating of 50 or less when tested in accordance with ASTM E 84.

2.1.1 Fire/Smoke Resistance Ratings

Where indicated, provide expansion control systems with fire/smoke barriers identical to those of systems tested for fire resistance per UL 2079 or ASTM E 1966 by a testing and inspecting agency acceptable to authorities having jurisdiction.

2.2 INTERIOR EXPANSION CONTROL SYSTEMS

2.2.1 Source Limitations

Obtain expansion control systems from single source from single manufacturer.

Silicone coating to be low-modulus, closed-cell, waterproof silicone applied to the impregnated foam sealant at a width greater than maximum allowable joint extension and which when cured and compressed will form a bellows. Product must be proved to have been tested and UL/ULC certified by Underwriters Laboratories to the joint-cycling and fire testing standards of UL 2079 where fire-resistant ratings are required.

The multi-component expansion joint system is to be installed slightly recessed from the surface such that when the field-applied injection band of silicone is applied at the installation side between the substrates and the foam-and-silicone-bellows, the system will be essentially flush with the substrate surface. Material must be capable of movements of +25%, -25% (50% total) of nominal material size.

Manufacturer approved-90-degree transition units must be used to transition joints into horizontal plane surfaces. Factory-fabricated units to be field-joined to straight lengths in accordance with manufacturer's prescribed methods and supplied joining materials.

2.2.2 Floor-to-Wall

Assembly consisting of compressed foam system with silicone bellow. Systems must allow unrestricted movement of joint without disengagement.

a. Design Criteria

- 1) Nominal Joint Width: As indicated on Drawings.
- 2) Minimum Joint Width: As indicated on Drawings.
- 3) Maximum Joint Width: As indicated on Drawings.
- 4) Movement Capability: +/-25 percent of total nominal joint width.
- 5) Type of Movement: Seismic.
- 6) Type: Compressed Foam.

- (a) Seal Material: Smoke/fire rated seismic pre-compressed horizontal foam system with silicone bellow.

Color: Gray.

2.2.3 Wall-to-Wall

Assembly consisting of compressed foam system with silicone bellow. Systems must allow unrestricted movement of joint without disengagement.

a. Design Criteria

- 1) Nominal Joint Width: As indicated on Drawings.
- 2) Minimum Joint Width: As indicated on Drawings.
- 3) Maximum Joint Width: As indicated on Drawings.
- 4) Movement Capability: +/-25 percent of total nominal joint width.
- 5) Type of Movement: Seismic.
- 6) Type: Compressed Foam.

- (a) Seal Material: Smoke/fire rated seismic pre-compressed horizontal foam system with silicone bellow.

Color: Gray.

2.2.4 Aluminum Joint Covers

ASTM E1399/E1399M. Provide aluminum floor-to-wall and floor-to-floor joint covers as indicated on drawings.

2.3 MATERIALS

2.3.1 Compression Seals

ASTM E1612; preformed elastomeric extrusions having an internal baffle system and designed to function under compression.

2.3.2 Cellular Foam Seals

Extruded, compressible foam designed to function under compression.

2.3.3 Fire Barriers

Any material or material combination, when fire tested after cycling, designated to resist the passage of flame and hot gases through a movement joint and to meet performance criteria for required fire-resistance rating.

2.3.4 Accessories

Manufacturer's standard anchors, clips, fasteners, set screws, spacers, and other accessories compatible with material in contact, as indicated or required for complete installations.

2.4 GENERAL FINISH REQUIREMENTS

- a. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 EXECUTION

3.1 EXAMINATION

Examine surfaces where expansion control systems will be installed for installation tolerances and other conditions affecting performance of work.

- a. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- a. Prepare substrates according to expansion control system manufacturer's written instructions.
- b. Coordinate and furnish anchorages, setting drawings, and instructions for installing expansion control systems. Provide fasteners of metal, type, and size to suit type of construction indicated and to provide for secure attachment of expansion control systems.

3.3 INSTALLATION

Comply with manufacturer's written instructions for storing, handling, and installing expansion control systems and materials unless more stringent requirements are indicated.

3.3.1 Compression Seals

Apply adhesive or lubricant adhesive as recommended by manufacturer to both frame interfaces or sides of slabs before installing compression seals.

3.3.2 Foam Seals

Install with adhesive recommended by manufacturer.

3.3.3 Termination Devices

Terminate exposed ends of expansion control systems with field- or factory-fabricated termination devices/sealants.

3.3.4 Fire/Smoke-Resistance-Rated Assemblies

Coordinate installation of expansion control system materials and associated work so complete assemblies comply with assembly performance requirements.

- a. Fire/Smoke Barriers: Install fire/smoke barriers to provide continuous, uninterrupted fire/smoke resistance throughout length of joint, including transitions and field splices.

3.3.5 Aluminum Joint Covers

Install per manufacturer's written instruction. At historic masonry fasten into mortar joints.

3.4 PROTECTION

- a. Do not remove protective covering until finish work in adjacent areas is complete. When protective covering is removed, clean exposed metal surfaces to comply with manufacturer's written instructions.
- b. Protect the installation from damage by work of other Sections. Where necessary due to heavy construction traffic, remove and properly store cover plates or seals and install temporary protection over expansion control systems. Reinstall cover plates or seals prior to Substantial Completion of the Work.

-- End of Section --